

Speech Emotion Recognition Using Deep Learning

Deep Learning-Based Human Emotion Classification from Speech Signals

Project Overview

Speech Emotion Recognition (SER) is a branch of Artificial Intelligence that enables computers to automatically recognize human emotions from spoken language. By analyzing acoustic characteristics of speech, the system predicts the emotional state of a speaker.

This project develops a complete Speech Emotion Recognition system using Deep Learning and Speech Signal Processing techniques. Mel-Frequency Cepstral Coefficients (MFCCs) are extracted from speech recordings and used to train Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) models for multi-class emotion classification.

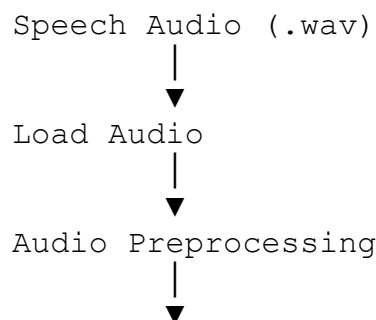
The trained model is deployed through a Streamlit web application, allowing users to upload speech recordings and receive real-time emotion predictions.

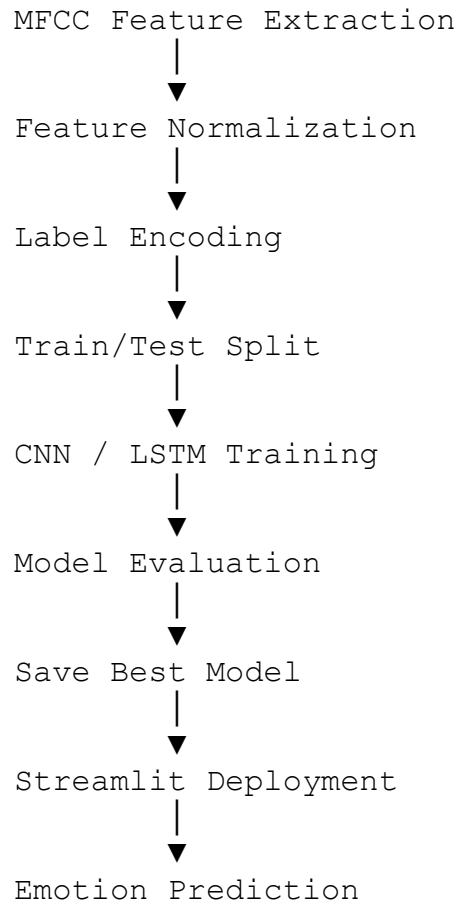
Objectives

The objectives of this project are to:

- Develop an intelligent Speech Emotion Recognition system.
 - Extract meaningful acoustic features from speech recordings.
 - Train Deep Learning models for emotion classification.
 - Compare CNN and LSTM architectures.
 - Evaluate model performance using standard classification metrics.
 - Deploy the trained model through a user-friendly Streamlit interface.
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Project Workflow





Dataset

RAVDESS Dataset

The project uses the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS), one of the most widely used benchmark datasets for Speech Emotion Recognition.

Dataset Information

- 24 professional actors
- Male and female speakers
- High-quality WAV recordings
- 8 emotion classes
- Speech and song recordings

Emotion Classes

Code Emotion

01 Neutral

Code Emotion

- 02 Calm
- 03 Happy
- 04 Sad
- 05 Angry
- 06 Fearful
- 07 Disgust
- 08 Surprised

Dataset Location

`/kaggle/input/ravdess-emotional-speech-audio/`

Feature Extraction

The project extracts Mel-Frequency Cepstral Coefficients (MFCCs), which are among the most effective features for speech analysis.

Why MFCC?

MFCCs capture important characteristics of the human vocal tract while reducing background variability.

For each audio recording:

- Audio is loaded using Librosa
 - Sampling rate preserved
 - First 40 MFCC coefficients extracted
 - Mean value computed across time
 - Final feature vector size = 40
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Data Preprocessing

The preprocessing pipeline includes:

1. Feature Scaling

MFCC features are standardized using StandardScaler.

Benefits:

- Faster convergence
- Stable training
- Better generalization

Saved as

```
scaler.joblib
```

2. Label Encoding

Emotion labels are converted into numerical values using LabelEncoder.

Example

Happy → 2

Sad → 4

Angry → 0

Saved as

```
label_encoder.joblib
```

3. Dataset Splitting

The dataset is divided into:

- Training Set: 80%
- Testing Set: 20%

Stratified sampling preserves class balance.

4. Feature Visualization

t-Distributed Stochastic Neighbor Embedding (t-SNE) is used to visualize high-dimensional MFCC features in two dimensions, helping assess emotion cluster separability.

Deep Learning Models

Two deep learning architectures were implemented and evaluated.

CNN Model

The CNN captures local feature patterns within MFCC vectors.

Architecture

Input Layer



Conv1D (64 filters)



MaxPooling1D



Dropout



Conv1D (128 filters)



MaxPooling1D



Flatten



Dense (128)



Dropout



Softmax Output

Advantages

- Learns local spectral features
 - Fast training
 - Good generalization
 - Lower computational cost
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LSTM Model

LSTM networks learn temporal dependencies in sequential data.

Architecture

Input Layer

↓

LSTM (128)

↓

Dropout

↓

LSTM (64)

↓

Dense (64)

↓

Dropout

↓

Softmax Output

Advantages

- Captures temporal information
- Suitable for sequential speech signals
- Better long-range dependency learning

Model Training

The models were trained using:

- Optimizer: Adam
- Loss Function: Sparse Categorical Crossentropy
- Batch Size: 32
- Epochs: 50

Training Callbacks

EarlyStopping

Stops training when validation loss stops improving.

Benefits

- Prevents overfitting
- Saves training time

ModelCheckpoint

Automatically saves the best-performing model.

Saved file

```
best_cnn_model.keras
```

Model Evaluation

The trained models are evaluated using:

Performance Metrics

- Accuracy
- Precision
- Recall
- F1 Score

Visualization

- Confusion Matrix
- Classification Report

- Training Accuracy Curve
 - Validation Accuracy Curve
 - Training Loss Curve
 - Validation Loss Curve
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Streamlit Web Application

A Streamlit web application was developed for real-time emotion recognition.

Features

- Upload WAV audio files
 - Automatic MFCC extraction
 - Emotion prediction
 - Prediction confidence scores
 - Interactive and user-friendly interface
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Project Structure

Speech-Emotion-Recognition/

```
|
|— dataset/
|
|— models/
|   |— best_cnn_model.keras
|   |— scaler.joblib
|   |— label_encoder.joblib
|
|— app.py
|— train.py
|— predict.py
|— feature_extraction.py
|— requirements.txt
|— README.md
|
|— notebooks/
```

Exported Files

File	Description
best_cnn_model.keras	Trained CNN model
scaler.joblib	Feature scaler
label_encoder.joblib	Label encoder
X_normalized.npy	Normalized feature matrix
y_encoded.npy	Encoded labels
app.py	Streamlit application
requirements.txt	Project dependencies

Installation

Clone the repository

```
git clone https://github.com/yourusername/Speech-Emotion-Recognition.git
```

```
cd Speech-Emotion-Recognition
```

Install dependencies

```
pip install -r requirements.txt
```

Running the Application

Start the Streamlit server

```
streamlit run app.py
```

If running in Google Colab

```
streamlit run app.py & npx localtunnel --port 8501
```

Open the generated URL in your browser.

Technologies Used

- Python
- TensorFlow / Keras
- Librosa
- NumPy
- Pandas

- Scikit-learn
 - Matplotlib
 - Streamlit
 - Joblib
-

Future Improvements

Potential enhancements include:

- Data augmentation using pitch shifting, time stretching, and noise injection.
 - Feature fusion with Chroma, Mel Spectrogram, Spectral Contrast, Zero Crossing Rate, and RMS Energy.
 - Transformer-based speech models.
 - Real-time microphone emotion detection.
 - Multilingual emotion recognition.
 - Deployment using Docker and cloud platforms.
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Conclusion

This project demonstrates a complete end-to-end Speech Emotion Recognition pipeline, from audio preprocessing and MFCC feature extraction to deep learning model training, evaluation, and deployment. By leveraging CNN and LSTM architectures, the system effectively classifies speech into multiple emotional categories and provides an interactive interface for real-time inference. The modular design also makes it suitable for further research, optimization, and integration into intelligent human-computer interaction systems.

Author

Name: JAM DIGITALS

Project: Speech Emotion Recognition Using Deep Learning

Tech Stack: Python, TensorFlow, Librosa, Scikit-learn, Streamlit

Purpose: Academic Project / AI Portfolio / CodeAlpha Internship